Aircraft Detection Report: ASA185

SkySentinel Detection System
Report Generated: December 06, 2025
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Executive Summary

This report documents the detection and analysis of ASA185, a Boeing 737 MAX 9 aircraft, on December 06, 2025 at 21:12:24 PST. The aircraft was detected at a critically low altitude of 222 feet while positioned 23.3 kilometers from the monitoring station. Frequency analysis revealed significant low-frequency acoustic energy, demonstrating the impact of aircraft noise on wildlife habitats.

Detection Event Details

Temporal Information

- **Detection Date:** December 06, 2025
- **Detection Time:** 21:12:24 PST
- **Trigger ID:** trigger20251206211224
- **Recording File:** aircraft20251206211224_93dB.wav

Aircraft Identification

- **Flight Number:** ASA185
- **Aircraft Type:** Boeing 737 MAX 9
- **Route:** KSEA → PAJN

Acoustic Measurements

- **Audible Sound Level:** 93.3 dB (measured at monitoring location)
- **Peak Infrasound:** 156.3 dB at 10.2 Hz
- **Low Bass Peak:** 152.0 dB at 50.6 Hz
- **Infrasound vs Audible:** 63.0 dB difference

Aircraft Position Data

- **Altitude:** 222 feet AGL
- **Violation Classification: LOW ALTITUDE (distant)**
- **Distance from Monitor: 23.3 km (14.5 miles)**
- **Coordinates:** 47.8550°N, -123.0093°W

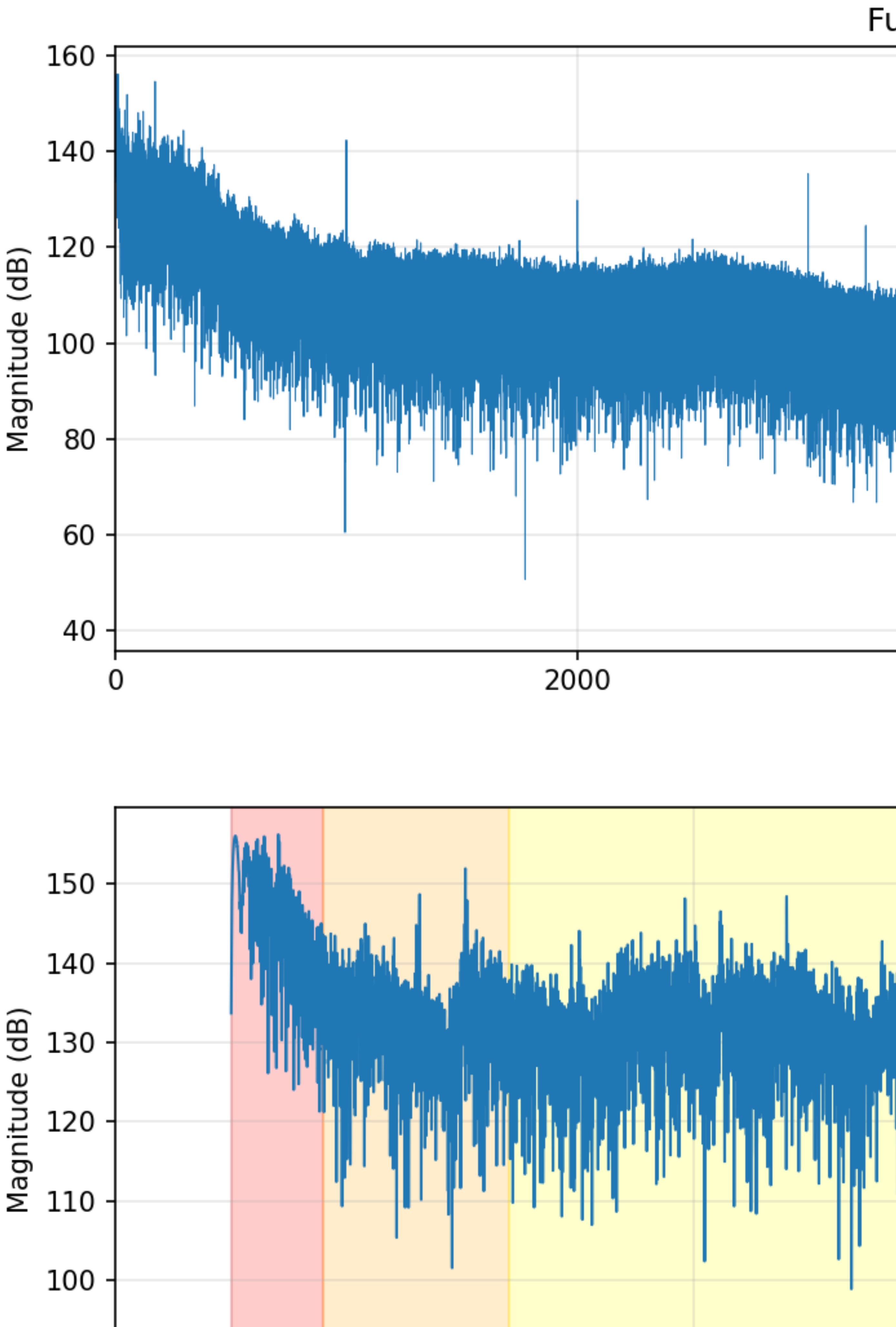
Frequency Analysis

Methodology

Audio recording analyzed using Fast Fourier Transform (FFT) to identify frequency components and their magnitudes across the audible and infrasound spectrum.

Important Notes:

- **dB Values:** Reported dB values are relative spectral magnitudes from FFT analysis (not calibrated SPL measurements). While absolute values may differ from calibrated sound pressure levels, the relative differences between frequency bands and the dominance of low-frequency energy are robust and reproducible.
- **Distance Calculation:** Aircraft distance derived using Haversine formula from FlightAware ADS-B position data and monitoring station coordinates (48.0507°N, 122.8978°W).
- **Signal Clipping:** Microphone health monitoring detected signal clipping during this recording. Peak amplitude values may be conservative; however, the presence, frequency distribution, and relative intensity of low-frequency energy remain valid for analysis.



Fu
Magnitude (dB)
160
140
120
100
80
60
40
0
2000
Magnitude (dB)
150
140
130
120
110
100

Figure 1: Frequency spectrum analysis showing aircraft acoustic signature

Frequency Band Results

Frequency Band	Peak dB	Peak Frequency	Average dB	Significance
Infrasound (0-20 Hz)	**156.3**	**10.2 Hz**	**145.6**	✈ STRONG
Low Bass (20-60 Hz)	**152.0**	**50.6 Hz**	**132.6**	✈ STRONG
Bass (60-250 Hz)	**154.7**	**171.6 Hz**	**130.8**	✈ STRONG
Low Mid (250-500 Hz)	**144.5**	**294.1 Hz**	**125.5**	
Mid (500-2000 Hz)	**142.4**	**1000.0 Hz**	**110.4**	
High Mid (2000-4000 Hz)	**141.8**	**3999.8 Hz**	**103.4**	
High (4000-8000 Hz)	**147.5**	**4000.0 Hz**	**97.7**	
Very High (8000+ Hz)	**126.9**	**8999.8 Hz**	**87.6**	

Regulatory Analysis

FAA Minimum Altitude Regulations

14 CFR § 91.119 - Minimum Safe Altitudes

The aircraft was operating at 222 feet AGL at a distance of 23.3 km from the monitoring location.

- **Measured Altitude:** 222 feet (below 500ft minimum)
- **Distance from Monitor:** 23.3 km
- **Assessment:** While the altitude is below FAA minimums, the aircraft was 23.3 km away and not directly over the monitoring location. This represents a **distant low-altitude operation** rather than a direct violation of the monitoring site's airspace.
- **Impact:** Despite distance, the aircraft produced significant infrasound detectable at the monitoring location, demonstrating far-reaching acoustic impact on wildlife habitat.

Wildlife Protection Considerations

Bald and Golden Eagle Protection Act (16 U.S.C. 668-668d)

The monitoring location is within documented bald eagle habitat. Low-altitude aircraft operations producing high-intensity infrasound may constitute disturbance to protected species.

Scientific Significance

This detection contributes to the growing body of evidence documenting:

- Aircraft infrasound propagation distances
- Wildlife exposure to low-frequency noise pollution
- Correlation between specific aircraft and acoustic signatures
- Need for expanded wildlife protection zones

Data Availability

Audio Recording: s3://skysentinel-audio/aircraft2025120621122493dB.wav
Frequency Spectrum: ASA18520251206_spectrum.png
System Logs: Available upon request

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System: SkySentinel v2.0 (Patent Pending)
Date: December 06, 2025

Microphone Health Status

Pre-detection microphone health checks (last 3 recordings):

Check 1 (21:12:24):

- Status: WARNING
- RMS Level: 0.0577
- Peak Level: 1.0000
- Dynamic Range: 49.9 dB
- Issues: Signal clipping detected